UNIVERSITY AT ALBANY
State University of New York

College of Arts and Sciences
Department of Mathematics and Statistics

Colloquium

Milen Yakimov
Northeastern University

Automorphism Groups, Dynamical Systems and Irreducible Representations – Faces of Cluster Algebras

Tuesday, October 30, 2026
3:00 p.m. Massry B010
(tea & coffee at 2:40 p.m.)

Abstract. A central problem in algebra is to describe the symmetries (automorphism groups) of algebraic structures. In dynamical systems, key problems are the descriptions of the phase spaces of dynamical systems (symplectic foliations of Poisson manifolds) and their global behaviour. In representation theory, one of the main problems is to describe the irreducible representations of quantum groups at roots of unity. This talk will be a gentle introduction to cluster algebra, invented by Fomin and Zelevinsky 25 years ago, through the lenses of these problems. Although the problems are quite distant from each other, we will witness that cluster algebras provide strong methods to tackle important instances of all three problems. The talk will be accessible to general audience and will assume no knowledge beyond linear algebra and the basics of manifolds.